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First-Round Questions to the Candidate

Paper 5

Memory Rank and Liftability of Transition Families Without a Generator: A Projection Theory for the Indivisibility Deficit

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Identifiers:	Q-P05-01 to Q-P05-18 (technical report); Q-PF-03, Q-PF-04, Q-PF-09 (portfolio-level evaluation)
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This document belongs to the examination record (03_examiner_questions). It lists the questions put to the candidate in the first round of the examination of Paper 5 and gives each a stable identifier, by which the Report, the Technical Annex and the correspondence can cite it.

1 Sources and identifiers

Technical report. Section 2 transcribes Section 7, "Questions to the author", of the Doctoral Examiner's technical report on Paper 5, `TR-P05-DOCv1-EX01-Rv1.pdf`. The report identifies the version examined as SSRN, DOI 10.2139/ssrn.71197798, 30 July 2026, and carries the evaluation date 4 September 2026. That identifier is a mistyped form of 10.2139/ssrn.7119779, and the text examined is, at every checkable point, that of 14 July 2026, not the later July deposit then current (Technical Annex, Section T.2(a)). Questions 3–6, 8 and 9 concern Section 8 of that text and its Table 1; the version of record withdraws the numerical section. The Doctoral Examiner sent the report to the candidate on 14 September 2026 (COR-P05-02), and the candidate answered, to both examiners, on 17 September 2026 (COR-P05-03).

Identifiers. `Q-P05-`k is Question k of that report. The candidate's letters and the Technical Annex cite the questions by these numbers. The text is the report's, word for word; its section signs (§) refer to sections of the paper, and its finding numbers to findings of the report. Notes in small type are not part of the questions. They identify unresolved references, later corrections and related questions.

Independent evaluation. The independent evaluator's review of Paper 5 (`TR-P05-DOCv1-EX02-Rv1.pdf`, Exhibit J, 02_technical_evaluations) states its points as comments and poses no questions. It was sent to the candidate on 11 September 2026 and is filed as COR-P05-01, the same

message as COR-P04-01; the candidate answered it together with the technical report on 17 September 2026 (COR-P05-03). It is not transcribed here.

Portfolio-level evaluation. Section 3 transcribes the questions of the portfolio-level technical evaluation (version 2, 4 September 2026; Exhibit H, Section 7, 02_technical_evaluations) that concern Paper 5. `Q-PF-`k is Question k of that evaluation and has the same identifier in all six documents; Q-PF-09 concerns all six papers and appears in each. The evaluation names the works W1–W6, and Wn is Paper n.

2 Questions of the technical report

Source: `TR-P05-DOCv1-EX01-Rv1`, Section 7, Questions 1–18.

Q-P05-01 Your model sets are closures. Prove that $V \in F_d$ implies an exact d-state lift in Θ_d^+, or rename the invariant. Consider the flat family generated by $M = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 0 \end{smallmatrix}\right)$: it has $h_2 = 0$ and no lift in Θ_2^+. Is the gap confined to families with degenerate one-time marginals?

Q-P05-02 Characterise a boundary sequence whose block masses tend to zero. Which conditional columns remain identifiable in the limit, and what statistical meaning attaches to $h_d = 0$ attained only through such a sequence?

Q-P05-03 Why does an optimiser value $h_3 \leq 0.026680$ not prove $h_3 > 0$? What form of certificate would prove non-membership in F_3, and can you produce one in the smallest nontrivial case $n = 2$, $d = 3$?

Q-P05-04 For every claimed zero in Table 1, exhibit the parameters. For a lumped chain the generating chain is such a witness, so the zeros need no optimisation at all — and, by Proposition 4.4, the witness pads to every higher dimension, which disposes of the h_5 entry.

Q-P05-05 Table 1 reports $h_4 = 0.000000$ and $h_5 \leq 0.000222$, which Proposition 4.4 forbids. What certifies the $d = 4$ entry if the $d = 5$ entry is wrong, and what tolerance should accompany each cell?

Q-P05-06 Apply your own Proposition 4.4 to split a hidden state of a three-state chain and lump the two copies together. The observed family is unchanged and its memory rank is three while the generating dimension is four. Restate the claim of the abstract and of §8.1 accordingly.

Q-P05-07 Does $\kappa_c(V) = 0$ imply that V is process-induced? I record a family with $\kappa_c = 0$ exactly whose pairwise-compatibility linear program is infeasible. What are the missing conditions, and would the linear-programming defect $\delta(V)$ — which vanishes on every exact compression and is Lipschitz in V by your own argument — not give a strictly stronger version of Theorem 5.6?

Q-P05-08 For the family of §8.2, what is κ_c? Theorem 5.6 is invoked for it without its hypothesis being checked.

Q-P05-09 Print the generating matrices and initial laws of the three lumped families of §8, or a seed and the exact generating procedure, and archive the raw optimiser record.

Q-P05-10 State the assumptions under which Moreau's decomposition applies. Is the Bouligand tangent cone of a semialgebraic model set necessarily convex, and what is the correct boundary statement?

Q-P05-11 Repair the bound of Proposition 7.1 when π_0 and π_1 define different weighted norms.

Q-P05-12 What exactly does Corollary 3.4 prove, and what further structure would be needed before speaking of instability, a categorical quotient, or a ring of invariants?

Q-P05-13 How does the memory rank relate to the minimal order of a hidden Markov model realizing the full path law, to nonnegative rank, and to Hankel rank? Which implications hold and which fail?

Q-P05-14 How does the liftability profile relate to the approximation of a process by hidden Markov models of assigned complexity, and to the computation of distances to the stochastic part of a model variety? What does your formulation add?

Q-P05-15 In the linear category the analogous problem is solved exactly by Adamyan–Arov–Krein theory. What is the obstruction in the stochastic category, and is it the same positivity obstruction that extension complexity describes?

Q-P05-16 Formulate the coherence analogy as an actual complex. Does δ_V square to zero on higher cochains?

Q-P05-17 What statistical assumptions would make the integer memory rank consistently estimable from noisy estimates of V, as distinct from estimating h_d?

Q-P05-18 Can the memory rank decrease under refinement of the time grid? Is there any monotonicity theorem in the grid?

3 Questions of the portfolio-level evaluation

Source: portfolio-level technical evaluation, version 2, 4 September 2026 (Exhibit H, 02_technical_evaluations), Section 7. Questions 3 and 4 concern Paper 5; Question 9 concerns all six papers.

Q-PF-03 **W5 §8.** Supply a lower bound certifying $h_3 > 0$ for the lumped-4→2 family — the route through your own Theorem 5.6 is cheapest — and account for the reported $h_5 = 0.000222$ exceeding $h_4 = 0.000000$ in violation of your Proposition 4.4.

Note. Q-P05-03 and Q-P05-05 ask the same questions. The evaluation, like the technical report, describes the text of 14 July 2026 (Technical Annex, Section T.2(a)).

Q-PF-04 **W5 reproducibility.** Bring §8 to the standard of W4 §9 and W6 §7: deposit the code and generating parameters, or replace the computations with exact arithmetic as W6 does.

Note. Q-P05-09 asks for the generating parameters and the raw optimiser record.

Q-PF-09 **Deposit venue.** State whether the works will be deposited on arXiv (math-ph / math.PR / quant-ph).